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Abstract

Under the new capital accord stress tests are to be included in market risk regulatory capital calculations. This development necessitates a coherent and objective framework for stress testing portfolios exposed to market risk. Following recent criticism of stress testing methods our tests are conducted in the context of risk models, building on the VaR literature. First, to identify the most suitable risk models for stress testing, we apply an extensive back testing procedure that focuses on extreme market movements. We consider eight possible risk models including both conditional and unconditional models and four possible return distributions (normal, Student's t , empirical and normal mixture) applied to three heavily traded currency pairs using a sample of daily data spanning more than 20 years. Finding that risk models accommodating both volatility clustering and heavy tails are the most accurate predictors of extreme returns, we develop a corresponding model-based stress testing methodology. Our results are compared with traditional stress tests and we assess the implications for capital adequacy. On the basis of our results we conclude that the new recommendations for market risk regulatory capital calculation will have little impact on current levels of foreign exchange regulatory capital.

Keywords: Value-at-Risk models, stress testing, market risk, exchange rates, GARCH

JEL classification: G18, G19, G21

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1. Introduction

A stress test is a risk management tool used to evaluate the potential impact on portfolio values of unlikely, although plausible, events or movements in a set of financial variables (Lopez, 2005). Stress tests are designed to explore the tails of the distribution of losses beyond the threshold (typically 99%) used in Value-at-Risk (VaR) analysis. They provide two vital pieces of information: the extent of potential losses in catastrophic circumstances and the scenarios in which such losses might occur. Such information is an input to decisions concerning, amongst other things, hedging, limit setting, portfolio allocations and capital adequacy.

Recent statements from the Basel Committee on Supervision suggest a new emphasis on stress testing. Since the end of 1997 financial institutions using internal VaR models to assess capital adequacy have been required to implement stress testing (see Basel Committee on Banking Supervision, 1996). While stress testing has been an input to capital adequacy since that time, the link has now been made more direct, i.e. 'A bank must ensure that it has sufficient capital to... *cover the results of its stress testing*' (Basel Committee on Bank Supervision (2006), at paragraph 778 (iii), p. 218, emphasis added). As a result leading industry practitioners have called for a re-examination of stress testing methodologies (see Rowe (2005a) and Rowe (2005b)).

A recent survey of stress testing practice (Committee on the Global Financial System, 2005) shows that most stress tests are currently designed around a series of scenarios based either on historical events, hypothetical events, or some combination of the two. They are typically selected by supervisors, directors or risk managers to reflect the particular portfolio composition or to reflect the risks perceived to be most pressing in the current environment. These methods have been criticised by Berkowitz (1999) and Greenspan (2000) for their lack of rigour, as they are typically conducted without a risk model so the probability of each scenario is unknown, making its importance difficult to evaluate. There is also a distinct possibility that many extreme yet plausible scenarios are not even considered.

Boyer, Gibson and Loretan (1999) argue that stress tests should be based on an appropriate model for the data generation process of the portfolio's returns and that stress tests conducted in the context of that model can provide a useful alternative or complement to the current ad hoc methods of stress testing. Several authors have attempted to build such a bridge between stress tests and risk models. The StressVaR method, named by Kupiec (1998) places stress tests in the context of the analytical VaR approach. The advantage of StressVaR is that the analysis takes account of links between assets, rather than focussing

on a single asset in isolation. Others studies including Kim and Finger (2000), Aragonés, Blanco and Dowd (2001) and Tan and Chan (2003) have developed stress tests for specific contexts such as correlation breakdown based on a variety of assumptions.

Our research takes a somewhat different direction to the papers cited above by focussing first on the modelling priorities that have emerged from the VaR literature since the mid-1990s and then designing model-specific stress tests. A first priority is the need to incorporate volatility clustering, as a market shock is likely to be followed by further large market movements on subsequent days. Another priority is to incorporate heavy tails and skewness, as the probability of extreme losses is greater than would be predicted under the assumption of normality, even allowing for changes in conditional volatility.

This paper explores the following questions:

1. Which risk models might be considered as candidates for ‘hosting’ a stress test?
2. How can stress tests be conducted in the context of these risk models and what would be the advantages and disadvantages of so doing?
3. Which risk models are empirically suitable for stress testing purposes?
4. What do the results of these stress tests imply for capital adequacy?

Our analysis differs from many previous VaR studies by first examining the ability of a wide variety of risk models to forecast extreme percentiles of returns distributions over multi-day horizons and then incorporating stress tests into only those models that pass our rigorous specification tests. Unlike many other studies we use assessment criteria based on expected tail loss (ETL) as well as value at risk (VaR). This procedure is designed to ensure that the model performs well for extreme outcomes beyond VaR, before the model is used for stress testing.

The outline of the paper is as follows: Section 2 explains the risk models that will be examined in this study, and the reasons for their selection. Section 3 addresses the question of how stress tests can be incorporated into risk models, giving specific examples. Section 4 presents an empirical analysis of extreme risk in currency markets using daily data since the mid-1970s (in the case of the GBP/USD and USD/JPY) and the mid-1980s (in the case of the AUD/USD). Section 5 evaluates stress tests in the context of the best performing empirical model and Section 6 summarizes and concludes.

2. Risk Models

What guidance can the VaR literature provide regarding the selection of appropriate risk models for stress testing? The tendency for market returns to exhibit volatility clusters has

always been a crucial issue for analysts attempting to understand large market movements. For example, the earliest widely known VaR models (see Morgan Guaranty Trust Company, 1996) incorporated volatility clustering, albeit in a crude form. The RiskMetrics Group popularised the Exponentially Weighted Moving Average (EWMA) method for estimating volatility, which is a constrained version of a GARCH model (see Alexander (2002) at Section 4.2.3).

Subsequently it became apparent that simple GARCH models with normal innovations are inadequate to explain extreme risk. The heavy tails apparent in financial return distributions led to the popularity of VaR methods based on historical simulation, which employs an empirical distribution (see Dowd and Rowe, 2004). Hull and White (1998) introduced a volatility weighted historical simulation methodology that adjusts historical portfolio returns to reflect changing conditional volatility.

More recently researchers have explored conditional non-normal innovation distributions for VaR modelling, finding them superior to their conditional normal counterparts. Mittnik and Paolella (2000) use an asymmetric generalised t distribution in combination with an A-PARCH model. McNeil and Frey (2000) use Extreme Value Theory (EVT) to estimate the tail of the residuals of a GARCH model. Angelidis, Benos and Degiannakis (2004) employ both t-GARCH and GED-GARCH, while So and Yu (2006) favour t-GARCH and Alexander and Lazar (2006) favour normal mixture GARCH.

This research, which typically explores the 99th percentile of outcomes at one-day horizons, suggests that accurate risk models will capture two key characteristics: volatility clustering and heavy tails. Our interest in stress testing requires exploration of both more extreme outcomes and, since immediate hedging may not be practical in a market crisis, longer horizons. Danielsson and Vries (2000) make the point that the most extreme market moves tend to exhibit reduced dependency between successive daily returns. This suggests that unconditional risk models may be sufficient for our application, provided they have sufficiently heavy tails.

We therefore examine a range of unconditional risk modelling approaches, each with its conditional counterpart, to give a total of eight risk models. Later in the paper we shall test the specification of these models using a single exposure only, so the models are specified in univariate form. For each model we shall estimate value at risk (VaR) and expected tail loss (ETL)¹ as shown in Table 1.

¹ ETL can also be described as Expected Shortfall or Conditional VaR. It is the expected loss conditional on exceeding VaR

Table 1: Calculation of ETL and VaR in Different Risk Models

Returns Distribution	1. Unconditional	2. Conditional
a. Normal	Analytic formulae	Monte Carlo simulation of normal GARCH process.
b. Empirical	Based on percentiles of a smoothed historical returns distribution	Monte Carlo simulation of GARCH process with innovations drawn from smoothed historical returns distribution
c. Student's t	Analytic formulae	Monte Carlo simulation of <i>t</i> -GARCH process.
d. Normal Mixture	Analytic formulae	Monte Carlo simulation of GARCH process with normal mixture innovations.

A recent study by Bams, Lehnert and Wolff (2005) has established that more complex tail modelling approaches come at the cost of greater uncertainty about the VaR estimates. Hence in selecting the eight models in Table 1 a guiding principle was to consider those that are relatively simple, accessible and parsimonious. We now describe each of the risk models, outlining their merits and explaining the calculation of VaR and ETL for each as a percentage of the current portfolio value.

1a. Unconditional Normal

This approach is included for the purpose of benchmarking the candidate risk models. Here we assume that $\varepsilon_t = y_t - \bar{y}$, where y_t , the daily log returns, are independent and normally distributed:

$$\varepsilon_t \sim N(0, \sigma^2) \quad (1.1)$$

Under this assumption the VaR and ETL for an horizon of h days at significance level α are:

$$\text{VaR}_{h,\alpha} = Z_\alpha \sigma_h \quad (1.2)$$

and

$$\text{ETL}_{h,\alpha} = \alpha^{-1} \varphi(Z_\alpha) \sigma_h \quad (1.3)$$

where Z_α is the absolute value of the standard normal critical value at $\alpha\%$, φ is the normal density function and $\sigma_h = \sigma\sqrt{h}$. Derivations of (1.3) and formulae for ETL for all unconditional risk models are provided in Appendix 1.

1b. Unconditional Empirical

This method is chosen by virtue of its popularity in the industry. Indeed, a recent survey found that 73% of banks that disclose their VaR methodology use historical simulation to calculate VaR (Perignon and Smith, 2006). It makes no assumption about the distribution of past returns, other than the assumption that returns are independent and identically distributed (i.i.d). If the empirical return distribution has heavy tails, then VaR measured at high percentiles will be greater than under the assumption of normality. This capacity to accommodate heavy tails is, no doubt, a large part of the attraction of this method.

This methodology can also accommodate skewed distributions as there is no assumption that the distribution of returns is symmetric. Thus VaR/ETL differ according to whether the portfolio is long or short the asset (in our case the commodity currency expressed in units of the terms currency). For the long asset case, VaR_α is -1 times the α percentile of the empirical return distribution and ETL_α is the absolute value of the average of all empirical return outcomes below VaR. Risk measures for the short asset case can be calculated using the same method, but first multiplying the return series by -1 .

The failings of historical simulation have been well described by Pritsker (2006). Of particular importance to this research is the fact that VaR at high percentiles is highly sample specific. Following Butler and Schachter (1998) we therefore smooth the empirical density by fitting the Epanechnikov kernel to a sample of daily returns. The daily VaR for a long portfolio is the return at the α percentile of the kernel density. To calculate ETL we take an average of tail VaRs as recommended by Dowd (2002) at p.45. That is, for a long portfolio VaR is calculated at successively lower percentiles (e.g. 0.99%, 0.98%, 0.97% ..) and ETL is a simple average of all these so-called tail VaRs.

To calculate VaR and the ETL over an h -day horizon we multiply the daily VaR or ETL by the square root of h . Our reasons for this are two-fold. First, Danielsson and Zigrand (2006) investigate the effect of jumps and non-zero drift on the scaling of VaR, finding that the square root of time rule underestimates VaR when jumps are possible but only at long horizons. Secondly, we have performed an empirical analysis that justifies the choice of square-root scaling. See Appendix 2 for details.

1c. Unconditional Student's t

One of the simplest ways to accommodate heavy tails is to assume they have a Student's t distribution with degrees of freedom parameter ν that is estimated using the method of moments. Under this risk model we assume the mean-adjusted returns are distributed:

$$\varepsilon_t \sim t_\nu \sigma \sqrt{(\nu - 2) \nu^{-1}} \quad (1.4)$$

where t_ν is the pdf for the Student's t with ν degrees of freedom, and σ is sample standard deviation. To calculate the $\alpha\%$ h -day VaR we follow the method described by Dowd (2002):

$$\text{VaR}_{h,\alpha} = T_{\alpha,\nu} \sigma_h \sqrt{\frac{\nu-2}{\nu}} \quad (1.5)$$

where $T_{\alpha,\nu}$ is the absolute critical value of the Student's t distribution at $\alpha\%$ for ν degrees of freedom. ETL assuming a Student's t distribution with density $f_\nu(x)$ is calculated as:

$$\text{ETL} = \alpha^{-1} (\nu-1)^{-1} (\nu-2 + X_\alpha^2) f_\nu(X_\alpha) \sigma_h \quad \text{where } X_\alpha = T_{\alpha,\nu} \sqrt{\frac{\nu-2}{\nu}}.$$

1d. Unconditional Normal Mixture

A normal mixture distribution is a probability-weighted combination of two (or more) normal distributions. Here we use two normal distributions: the first reflects 'typical' market conditions while the second, having higher volatility and lower probability, reflects stressed market conditions. Depending on the means and variances of the individual normal densities, the mixture density can exhibit the positive excess kurtosis and negative skew often observed in financial markets. These characteristics suggest that a normal mixture distribution is well-suited to stress testing.

The mixture density is defined as

$$g(\varepsilon) = \pi \varphi_1(\varepsilon) + (1-\pi) \varphi_2(\varepsilon) \quad (1.6)$$

where $\varphi_1(\varepsilon)$ and $\varphi_2(\varepsilon)$ are normal density functions, π is the probability weight on the first normal density having mean μ_1 and variance σ_1^2 , and $1-\pi$ is the weight on the second normal density having mean μ_2 and variance σ_2^2 . Note that while we have defined the overall mixture distribution to have zero mean, the two component densities can have different, non-zero means. It is this feature that creates the possibility of skewness in the mixture density and therefore risk measures can differ depending on whether one is long or short the currency.

We estimate the parameters from historical data using the EM algorithm. To determine the one-day VaR (expressed as a percentage of portfolio value) we solve the following expression for VaR_α .²

$$\pi \text{Prob}(Z < [\text{VaR}_\alpha - \mu_1]/\sigma_1) + (1-\pi) \text{Prob}(Z < [\text{VaR}_\alpha - \mu_2]/\sigma_2) = \alpha \quad (1.7)$$

² For portfolios with a short asset position the same method can be used, but the portfolio returns should first be multiplied by -1.

The ETL (derived in Appendix 1) is calculated using

$$\text{ETL}_\alpha = \alpha^{-1} \left[\pi N(-\sigma_1^{-1} \text{VaR}_\alpha) \sigma_1 + (1 - \pi) N(-\sigma_2^{-1} \text{VaR}_\alpha) \sigma_2 \right] - \pi \mu_1 - (1 - \pi) \mu_2 \quad (1.8)$$

Both VaR and ETL are scaled using the square root of time rule in the case of horizons beyond one day.

2a. Conditional Normal

Here the mean adjusted returns are assumed to be conditionally normally distributed with conditional variance following the symmetric GARCH(1,1) process of Bollerslev (1986):

$$\sigma_t^2 = \gamma_1 + \gamma_2 \varepsilon_{t-1}^2 + \gamma_3 \sigma_{t-1}^2 \quad \gamma_1 \geq 0, \gamma_2, \gamma_3 > 0, \gamma_2 + \gamma_3 < 1 \quad (1.9)$$

Use of an asymmetric GARCH model was considered but discarded for two reasons. First, the present study focuses on currency returns where symmetric GARCH models are typical (see Vilasuso (2002) and So and Yu (2006)). Second, we tested the data for asymmetry in conditional volatility and found no consistent evidence to support its presence.

To calculate VaR and ETL we simulate the process over an h -day horizon with 30,000 paths. For a long (short) position the $\alpha\%$ h -day VaR is the portfolio return calculated at the α ($1-\alpha$) percentile and the corresponding ETL is the mean of all simulated returns below (above) the VaR.

2b. Conditional Empirical

This method is adapted from the filtered historical simulation method of Barone-Adesi, Bourgoin and Giannopoulos (1998). First we fit a normal-GARCH model to historical data as for method 2a, then we standardize the returns by multiplying by the current conditional volatility estimate and dividing by the conditional volatility estimate corresponding to that return. Then we fit the Epanechnikov kernel to provide a smoothed distribution of standardised returns. We simulate the GARCH model forward over the h -day risk horizon using innovations that are sampled from the kernel density. The $\alpha\%$ h -day VaR for a long (short) asset is the absolute portfolio return calculated at the α ($1 - \alpha$) percentile and the corresponding ETL is the mean of all simulated returns below (above) the VaR.

2c. Conditional Student's t

We use a standard GARCH(1,1) specification for conditional variance as in 2a, but now the innovations are drawn from a Student's t distribution with ν degrees of freedom, in order to capture the excess kurtosis present in empirical data. This t -GARCH model is available

in many statistical packages, making it readily accessible to many risk analysts. While more complex heavy-tailed error specifications have been explored by some researchers we select t -GARCH on account of its relative accessibility and parsimony. The parameters are estimated over the historical sample period, then we simulate forward 30,000 paths over the h -day risk horizon. As in the other conditional models, for a long (short) position the $\alpha\%$ h -day VaR is the absolute portfolio return calculated at the α ($1 - \alpha$) percentile and the corresponding ETL is the mean of all simulated returns below (above) the VaR.

2d. Conditional Normal Mixture

We start by fitting a standard GARCH(1,1) model with normal innovations to historical returns, then standardise the historical returns in the same way as for 2b. Using the EM algorithm we fit a normal mixture distribution to the standardised returns having two component densities. To calculate VaR at the h -day horizon we simulate (1.9) over an h -day horizon, using innovations drawn from the normal mixture distribution. $\text{VaR}_{h,\alpha}$ for a long (short) position is the absolute portfolio return calculated at the α ($1-\alpha$) percentile and the corresponding ETL is the absolute mean of all simulated returns below (above) the VaR.

3. Stress Testing Methodology

A stress test measures likely losses resulting from an extreme market event. It can be thought of as a two-stage process: specifying the initial shock event, and then specifying the subsequent response to that shock which may extend over several hours or days. We present here a new approach to ‘model-based stress tests’ in which the risk model determines both the extreme market event and the after-effects of the shock. Alternatively, a hypothetical shock event could be selected and the model used only for analysing its consequences, but we do not consider such an approach here.

The ability to analyse the after-effects of a shock is an important advance on stress tests that consider only the instantaneous portfolio impact of an extreme event. This is particularly true when the analyst also considers the implications for market liquidity of the stress event, as required by regulators.

3.1 Initial Shock Event

We start by considering the initial shock event. This represents a major gap or discontinuity in price resulting from important unanticipated information entering the market. The traditional approach described in Committee on the Global Financial System (2005) has been to base this shock on an historical or hypothetical event specified by analysts,

management or regulators. An alternative or complementary approach is to consider an extreme outcome as defined under the risk model. An ad hoc rule used by many practitioners is to model the shock as a return that is ‘six sigma’ below the mean return, where sigma is the volatility of returns. But why six, and what is sigma? In the following we propose rules for defining the initial shock in each of the risk models considered. We assume here that portfolio returns are modelled directly. If, however, the portfolio positions have been mapped to one or more risk factors, then the shock could be applied instead to a selected risk factor.

Let α denote the probability of an extreme market event occurring on a given day. The value of α should be fixed at board level to reflect the risk profile of the organisation and, if relevant, its target credit rating.

- i. When the model assumes a normal distribution for returns we consider a shock ε^* equivalent to

$$\varphi^{-1}(1 - \alpha) \sigma \mathbf{1}$$

where φ is the standard normal density and $\mathbf{1}$ is an indicator variable taking the value -1 if we are long the asset and 1 otherwise. For example, if $\alpha = 0.0001$ and we are long the asset, then we consider an initial shock $\varepsilon^* = -3.719\sigma$.

- ii. Under the empirical method the initial shock ε^* for the long asset position is simply the α percentile of the empirical/kernel distribution (or the $(1 - \alpha)$ percentile for a short position).³
- iii. When the model assumes a Student’s t distribution for returns we use:

$$\varepsilon^* = f_{\nu}^{-1}(1 - \alpha) \sigma \sqrt{(\nu - 2)/\nu} \mathbf{1}$$

where f_{ν} is the t density with ν degrees of freedom.

- iv. When the model assumes a normal mixture distribution for returns:

$$\varepsilon^* = \varphi^{-1}(1 - \alpha(1 - \pi)^{-1}) \sigma_2 \mathbf{1}.$$

Notice that ε^* is therefore a result of two decisions: the choice of α and the distribution chosen to describe innovations. The long-term standard deviation σ is the sample standard deviation of a large sample of data; we use all available data (see Section 4).

³ A variation on this approach would be to use Extreme Value Theory. That is, the generalised Pareto distribution parameters (ξ and β) are estimated from the empirical data and used to calculate VaR_{α} , which becomes the initial shock in the stress test. See Hull (2006), Section 9.4, for further detail.

3.2 Modelling the After-Shock

Portfolio returns are assessed for $h - 1$ days after the initial shock so that the total stress test, including the initial shock, has a horizon of h days⁴. Having specified the initial shock event, we then assess the subsequent after-shock using a suitable risk model. The consequences of a shock event can include some or all of the following: further large moves in the same market (as volatility clustering predicts), large moves in other markets, higher correlations between markets, increased implied volatility in option markets, and reduced market liquidity. Risk models have been developed to address all of these issues, although in some cases the literature is still relatively sparse.⁵

For reasons of space this study only captures volatility clustering and a crude adjustment for liquidity when modelling the after-shock. For reasons that will be discussed in Section 5, we examine risk horizons of up to 20 trading days, allowing for both managerial delay in reacting to shocks and reduced market liquidity

Before examining the ‘after-shock’ risk models in detail, we must first consider the choice of sigma immediately prior to the stress event. That is, what is the risk context for the shock event? This will help determine the course of risk subsequent to the shock. For this purpose we propose calculating sample standard deviation using a long sample of data (σ) which will provide a relatively benign value for sigma.⁶ That is, the market prior to the stress test is assumed to have ‘typical’ risk characteristics.

(a) Unconditional Models

When returns are i.i.d. there is no reason to suppose that the after-shock returns will be more volatile than usual, so the after-shock standard deviation is the model’s long-term standard deviation σ . This governs returns for the $h - 1$ days following the initial stress event.

The outcome of the stress test is expressed as a range of possible values around the initial shock. This emphasizes the fact that if a trading portfolio is maintained on days following

4 The after-shock could, alternatively, be modelled using intra-day data in which case we would assess $h-1$ units of time, where a unit of time could be defined as, say, 5-minute time intervals.

5 Volatility clustering has been comprehensively studied in the vast GARCH literature, surveyed, for example, by Bollerslev, Engle and Nelson (1994). The links between markets (including changes in correlation) can best be examined in multivariate GARCH models, surveyed by Bauwens, Laurent and Rombouts (2004). Surprisingly little is known about modelling the vega risk of option portfolios, although Malz (2001), Lehar, Scheicher and Schittenkopf (2002) and Barone-Adesi, Giannopoulos and Vosper (2002) provide some possible solutions. The issue of market liquidity is of utmost importance to portfolio stress testing and its analysis is required by regulators. Traders hedging large positions after a shock event may find that they cannot transact efficiently, resulting in exposure to adverse market conditions for longer periods, wider spreads or adverse market impact. Models of market liquidity risk are emerging (see Malz (2003) for a survey, and Angelidis and Benos (2006) for a recent example), but they often focus on ‘typical’ market conditions rather than crisis periods

6 For the conditional models it would be possible to calculate long-run volatility using $\omega/(1-\alpha-\beta)$.

the initial stress event, there is a wide range of possible outcomes. Some are worse than the initial shock while some are considerably better as the portfolio benefits from market recovery. We therefore construct a confidence interval describing the likely range of stress test outcomes (expressed as a percentage of portfolio value) resulting from the shock event as follows:

$$[\text{VaR}_{p,h-1} - 1 \epsilon^*, \text{VaR}_{1-p,h-1} + 1 \epsilon^*]$$

where VaR is calculated according to the risk model described in Section 2, if necessary setting σ equal to long-term sample standard deviation. In the case of the normal mixture model, we use the long-term estimate of σ_2 in place of σ .

A simulated stress test for unconditional models can also be implemented, the advantage being that one can analyse how portfolio returns would be impacted by hedging strategies (whether gradual or immediate), the implications for limits and the likely margin calls for each path. Further details are available from the authors on request.

(b) Conditional Models

Here model-based stress tests are implemented using Monte Carlo simulation to reflect changing volatility over the life of the stress test. We assume that variance is equal to the long-term variance σ^2 on day 1 of the test but there is an extreme innovation on that day, as defined by the initial shock event above. On day 2 of the test, variance will increase significantly in response to the shock event. Adapting 1.9 the variance on day 2 is defined as:

$$\sigma_t^2 = \hat{\gamma}_1 + \hat{\gamma}_2 \varepsilon^{*2} + \hat{\gamma}_3 \sigma^2$$

Hereafter the simulation proceeds in the usual way, with innovations drawn from the prescribed distribution (normal, empirical/kernel, Student's t or normal mixture), scaled to the appropriate conditional variance, and applying 1.9 to determine the variance on subsequent days during the life of the stress test. We simulate 30,000 paths each with $h - 1$ realizations, aggregate the daily returns along each path to give the simulated h -day return under the assumption that the portfolio is held constant for the duration of the stress test.⁷ Note that for some paths the worst case portfolio outcome is not at the end of the risk horizon but at an interim point.

To summarise the output, we recommend presenting stress results as a likely range of possible stress test outcomes. The best case outcome for a long asset is defined as the

⁷ Alternatively we can consider dynamic adjustments to the portfolio during the stress test to compare alternative hedging strategies.

$(1 - \rho)$ percentile of the simulated h -day returns when ranked from highest to lowest and the worst case outcome is the ρ percentile of the simulated h -day returns.⁸

4. Model Selection

In this section we back-test our risk models using the Kupiec (1995) test for coverage, the Christoffersen (1998) test for conditional coverage and a method for back-testing ETL due to McNeil and Frey (2000). The empirical analysis here differs from previous empirical research in several respects, namely:

- To better reflect the stress testing context of this research we consider confidence levels for VaR/ETL of 99.0%, 99.5% and 99.9% (whereas most VaR research has analysed only the 99.0% confidence level). The accuracy of our risk measures is assessed using over 15,000 out-of-sample returns.
- To account for market liquidity constraints we consider a 3-day risk horizon in addition to the more typical 1-day horizon.
- The choice of sample period is an important source of model risk⁹. Thus we consider a range of possible sample periods (250, 500, 1000 and 2000 days) for estimating parameters, volatility and percentiles.

We evaluate the risk models in the context of daily returns on three of the most traded currency pairs for which a relatively long history is available: the Australian dollar in terms of US dollars (AUD/USD), the Great Britain Pound in terms of US dollars (GBP/USD) and the US dollar in terms of Japanese Yen (USD/JPY). According to Bank for International Settlements (2005), these were the three most actively traded currency pairs after the Euro/USD in 2004, and the Euro/USD could not be included in the analysis due to lack of historical data.¹⁰ We have 5,691 observations on AUD/GBP dating from the float of the AUD in December 1983 until June 2006 and 8,154 observations from January 1974 until June 2006 for the USD/JPY and GBP/USD.

For each series we repeatedly estimate both VaR and ETL for each of the eight risk models. The maximum sample size for estimation of parameters is 2000 days, so the first VaR/ETL estimation occurs 2000 days (nearly 8 years) into the data sample. In the case of the USD/JPY and the GBP/USD this allows us to calculate over 6,000 estimates of VaR/ETL having a 1-day horizon, or around 2,000 non-overlapping estimates of VaR/ETL having a 3-day

⁸ In the short asset case the same analysis is performed but path returns are first multiplied by -1.

⁹ See discussion of sample period in relation to historical simulation in Section 9.2 of Hull, John, 2006. *Risk Management and Financial Institutions* (Pearson Prentice Hall).

¹⁰ Data obtained from (www.rba.gov.au/Statistics/HistoricalExchangeRates/index.html) and (www.federalreserve.gov/releases/h10/Hist/default.htm).

horizon. Each time VaR and ETL are estimated, they are based on revised parameter estimates (or percentiles) using the most recent sample of 250, 500, 1000 or 2000 days.

At the end of each risk horizon we calculate actual profits/losses for the trading portfolio. An exceedance occurs if the loss is greater than the estimated VaR for that risk horizon. These exceedances are the inputs to the tests of coverage, conditional coverage and ETL referred to above. Tables 2-9 present results for each of the eight risk models using a sample period of 2000 days and we summarise the findings as follows:

- The assumption of normality of returns cannot be justified whether in the conditional or unconditional context. Column (c) of Table 2 shows that in almost every case we can comfortably reject the hypothesis that the actual number of exceptions is equal to the expected number of exceptions. The model performs no better in a test of conditional coverage (column (d)), and performance in measuring ETL is also poor. Column (e) shows that in every case we can reject the null hypothesis (with 95% confidence) that ETL does not consistently understate the true potential for losses beyond the VaR. Turning to the conditional normal risk model (Table 6), we observe that while modelling heteroscedasticity improves performance, especially in terms of conditional coverage, the results do not generally support the use of this risk model.

Table 2: Backtest of Unconditional Normal

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	0.00%	0.02%	0.00%
	99.5%	0.00%	0.00%	0.00%
	99.9%	0.00%	0.02%	0.00%
Short AUD/Long USD	99.0%	1.24%	4.24%	0.00%
	99.5%	0.00%	0.02%	0.00%
	99.9%	0.00%	0.00%	0.00%
Long GBP/Short USD	99.0%	0.00%	0.00%	0.00%
	99.5%	0.00%	0.00%	0.00%
	99.9%	0.00%	0.00%	0.50%
Short GBP/Long USD	99.0%	9.52%	6.30%	0.00%
	99.5%	5.44%	1.67%	0.00%
	99.9%	0.00%	0.00%	0.10%
Long USD/Short JPY	99.0%	0.00%	0.00%	0.00%
	99.5%	0.00%	0.00%	0.00%
	99.9%	0.00%	0.00%	0.00%
Short USD/Long JPY	99.0%	12.15%	17.81%	0.00%
	99.5%	0.24%	0.18%	0.00%
	99.9%	0.00%	0.00%	0.00%
3-day horizon				
Long AUD/Short USD	99.0%	0.06%	0.08%	0.30%
Short AUD/Long USD	99.0%	63.36%	75.96%	4.50%
Long GBP/Short USD	99.0%	0.00%	0.00%	0.00%
Short GBP/Long USD	99.0%	74.38%	46.75%	0.60%
Long USD/Short JPY	99.0%	0.00%	0.00%	0.00%
Short USD/Long JPY	99.0%	90.95%	81.58%	0.70%

- (a) The trading portfolio used for the analysis.
(b) The confidence level $(1-\alpha)$ at which VaR and ETL are calculated.
(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.
(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.
(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

Table 3: Backtest of Unconditional Empirical (with Kernel)

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	40.90%	56.93%	14.50%
	99.5%	56.04%	74.84%	5.80%
	99.9%	27.03%	53.94%	7.40%
Short AUD/Long USD	99.0%	50.51%	62.43%	23.70%
	99.5%	42.14%	21.86%	38.90%
	99.9%	5.23%	0.62%	70.30%
Long GBP/Short USD	99.0%	95.20%	8.83%	96.00%
	99.5%	74.74%	82.76%	94.80%
	99.9%	63.06%	88.72%	12.60%
Short GBP/Long USD	99.0%	1.80%	0.76%	73.10%
	99.5%	6.11%	3.01%	60.40%
	99.9%	15.75%	36.77%	3.90%
Long USD/Short JPY	99.0%	57.14%	0.26%	15.80%
	99.5%	45.39%	0.29%	22.20%
	99.9%	15.51%	1.43%	16.70%
Short USD/Long JPY	99.0%	20.97%	9.98%	54.00%
	99.5%	61.18%	26.08%	51.60%
	99.9%	95.06%	99.23%	28.10%
3-day horizon				
Long AUD/Short USD	99.0%	63.36%	75.96%	91.20%
Short AUD/Long USD	99.0%	9.84%	24.52%	71.50%
Long GBP/Short USD	99.0%	4.88%	3.03%	82.20%
Short GBP/Long USD	99.0%	4.06%	11.45%	4.60%
Long USD/Short JPY	99.0%	33.54%	0.72%	19.40%
Short USD/Long JPY	99.0%	12.54%	28.07%	38.30%

(a) The trading portfolio used for the analysis.

(b) The confidence level $(1-\alpha)$ at which VaR and ETL are calculated.

(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.

(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.

(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

- The risk models based on the empirical distribution are much more suited to stress testing, especially in the conditional case. Results for the unconditional empirical model are presented in Table 3. In the majority of cases considered, we cannot reject the hypotheses tested at 95% confidence. We note, however, that in some cases there is a tendency for exceptions to be clustered, so the hypothesis of no conditional coverage is rejected (at 95%) in 8 out of 24 cases. The conditional empirical risk model (described in Table 7) eliminates this problem.

Table 4: Backtest of Unconditional t

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	0.03%	0.04%	9.70%
	99.5%	1.35%	2.41%	6.20%
	99.9%	0.06%	0.27%	87.70%
Short AUD/Long USD	99.0%	61.28%	66.68%	10.80%
	99.5%	21.58%	16.37%	32.00%
	99.9%	27.03%	1.20%	68.30%
Long GBP/Short USD	99.0%	15.34%	1.58%	95.10%
	99.5%	20.73%	35.65%	99.70%
	99.9%	35.33%	64.83%	87.10%
Short GBP/Long USD	99.0%	5.21%	2.38%	24.40%
	99.5%	20.34%	9.97%	24.10%
	99.9%	63.06%	88.72%	37.30%
Long USD/Short JPY	99.0%	0.10%	0.00%	0.10%
	99.5%	0.00%	0.00%	10.10%
	99.9%	0.10%	0.05%	56.00%
Short USD/Long JPY	99.0%	7.14%	3.32%	56.60%
	99.5%	37.62%	17.54%	86.80%
	99.9%	35.33%	64.83%	75.50%
3-day horizon				
Long AUD/Short USD	99.0%	7.54%	11.92%	58.10%
Short AUD/Long USD	99.0%	49.57%	73.04%	91.80%
Long GBP/Short USD	99.0%	0.10%	0.18%	87.30%
Short GBP/Long USD	99.0%	7.40%	18.66%	3.80%
Long USD/Short JPY	99.0%	0.62%	0.02%	2.40%
Short USD/Long JPY	99.0%	7.40%	18.66%	48.80%

(a) The trading portfolio used for the analysis.

(b) The confidence level $(1-\alpha)$ at which VaR and ETL are calculated.

(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.

(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.

(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

- Use of the Student's t distribution may be justified for stress testing purposes in the conditional t case. Table 4 presents the unconditional t results which are mixed. The model is rejected (at 95% confidence) for the long AUD/USD and long USD/JPY portfolios. This can be explained by the negative skewness observed in both currency pairs. Column (d) also provides some evidence that this risk model is flawed by clustering of exceedances. Moving to the conditional t risk model (Table 8), we observe that in 16 of the 24 cases, we cannot reject the model (with 95% confidence) under the test of conditional coverage. In 21 out of 24 cases we cannot reject the ETL measure

at the 95% confidence level. Even at very low levels of α (high confidence levels) the measure of ETL under this risk measure conservatively estimates the potential for losses beyond the VaR. It could be argued, however, that the conditional t risk model is too conservative in some cases since no exceedances are recorded for two of the portfolios.

Table 5: Backtest of Unconditional Normal Mixture

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	2.74%	3.85%	0.00%
	99.5%	0.12%	0.32%	0.00%
	99.9%	0.00%	0.02%	2.10%
Short AUD/Long USD	99.0%	40.90%	56.93%	0.70%
	99.5%	14.71%	13.21%	0.10%
	99.9%	0.21%	0.07%	1.30%
Long GBP/Short USD	99.0%	4.27%	3.25%	1.40%
	99.5%	1.61%	3.97%	13.00%
	99.9%	15.51%	35.81%	3.30%
Short GBP/Long USD	99.0%	39.42%	18.21%	0.5%
	99.5%	88.96%	33.20%	0.1%
	99.9%	28.30%	55.46%	0.6%
Long USD/Short JPY	99.0%	2.37%	0.00%	0.00%
	99.5%	0.14%	0.00%	0.00%
	99.9%	0.00%	0.00%	0.30%
Short USD/Long JPY	99.0%	0.2%	0.44%	34.1%
	99.5%	9.5%	4.72%	47.6%
	99.9%	95.06%	99.23%	56.6%
3-day horizon				
Long AUD/Short USD	99.0%	20.29%	35.02%	5.30%
Short AUD/Long USD	99.0%	49.57%	73.04%	39.20%
Long GBP/Short USD	99.0%	0.05%	0.10%	3.70%
Short GBP/Long USD	99.0%	42.22%	62.86%	1.6%
Long USD/Short JPY	99.0%	3.04%	0.04%	0.00%
Short USD/Long JPY	99.0%	2.06%	6.46%	42.00%

- (a) The trading portfolio used for the analysis.
(b) The confidence level ($1-\alpha$) at which VaR and ETL are calculated.
(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.
(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.
(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

- Tables 5 and 9 indicate that the normal mixture distribution cannot be justified for stress testing purposes, despite its intuitive appeal. Even in the conditional case, this risk

model fails the test of conditional coverage in 7 of 24 cases (at 95% confidence) and fails the ETL test in 14 of 24 cases¹¹.

Table 6: Backtest of Conditional Normal

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	0.00%	0.02%	0.00%
	99.5%	0.00%	0.02%	0.00%
	99.9%	0.00%	0.02%	0.00%
Short AUD/Long USD	99.0%	2.74%	8.30%	0.50%
	99.5%	0.42%	1.26%	5.60%
	99.9%	5.23%	14.95%	6.00%
Long GBP/Short USD	99.0%	0.00%	0.02%	0.00%
	99.5%	0.00%	0.02%	0.00%
	99.9%	0.00%	0.02%	2.20%
Short GBP/Long USD	99.0%	7.37%	20.16%	0.10%
	99.5%	5.44%	11.78%	0.00%
	99.9%	0.10%	0.43%	0.00%
Long USD/Short JPY	99.0%	0.00%	0.00%	0.00%
	99.5%	0.00%	0.00%	0.00%
	99.9%	0.00%	0.00%	0.00%
Short USD/Long JPY	99.0%	23.63%	48.65%	0.00%
	99.5%	0.40%	1.10%	0.20%
	99.9%	0.00%	91.77%	0.10%
3-day horizon				
Long AUD/Short USD	99.0%	2.35%	5.34%	2.00%
Short AUD/Long USD	99.0%	9.84%	24.52%	98.10%
Long GBP/Short USD	99.0%	0.00%	0.02%	17.90%
Short GBP/Long USD	99.0%	16.96%	27.15%	0.90%
Long USD/Short JPY	99.0%	0.00%	0.00%	0.00%
Short USD/Long JPY	99.0%	90.95%	81.59%	5.60%

- (a) The trading portfolio used for the analysis.
(b) The confidence level (1- α) at which VaR and ETL are calculated.
(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.
(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.
(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

¹¹ We also assessed an alternative Conditional Normal Mixture specification (see Alexander, Carol, and Emese Lazar, 2006, Normal Mixture GARCH(1,1): Applications to Exchange Rate Modelling, *Journal of Applied Econometrics* 21, 307-336.) with similar results.

Table 7: Backtest of Conditional Empirical (with Kernel)

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	98.68%	68.73%	93.00%
	99.5%	27.79%	52.63%	58.70%
	99.9%	9.60%	25.02%	0.00%
Short AUD/Long USD	99.0%	51.12%	59.83%	52.10%
	99.5%	91.60%	91.04%	74.60%
	99.9%	71.02%	93.11%	62.20%
Long GBP/Short USD	99.0%	9.60%	16.88%	72.20%
	99.5%	9.50%	22.93%	48.80%
	99.9%	95.06%	99.23%	35.60%
Short GBP/Long USD	99.0%	55.67%	49.37%	86.80%
	99.5%	28.14%	50.55%	72.40%
	99.9%	63.06%	88.72%	89.70%
Long USD/Short JPY	99.0%	84.39%	54.32%	10.40%
	99.5%	68.97%	77.29%	13.90%
	99.9%	73.82%	93.82%	2.10%
Short USD/Long JPY	99.0%	94.58%	89.35%	68.00%
	99.5%	48.70%	21.83%	18.40%
	99.9%	47.66%	76.81%	21.50%
3-day horizon				
Long AUD/Short USD	99.0%	84.25%	85.32%	94.50%
Short AUD/Long USD	99.0%	9.84%	24.52%	99.00%
Long GBP/Short USD	99.0%	45.07%	42.44%	61.20%
Short GBP/Long USD	99.0%	58.78%	66.51%	35.60%
Long USD/Short JPY	99.0%	24.21%	36.13%	0.20%
Short USD/Long JPY	99.0%	73.44%	79.04%	13.50%

(a) The trading portfolio used for the analysis.

(b) The confidence level ($1-\alpha$) at which VaR and ETL are calculated.

(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.

(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.

(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

Table 8: Backtest of Conditional t

Sample Size = 2000 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	14.71%	11.44%	69.30%
	99.5%	30.66%	51.33%	92.20%
	99.9%	33.46%	62.71%	73.00%
Short AUD/Long USD	99.0%	17.45%	31.61%	99.70%
	99.5%	0.60%	2.25%	88.40%
	99.9%	NA (no exceedances)	NA (no exceedances)	NA (no exceedances)
Long GBP/Short USD	99.0%	74.41%	53.55%	96.10%
	99.5%	88.96%	85.50%	98.70%
	99.9%	15.75%	36.77%	86.60%
Short GBP/Long USD	99.0%	7.14%	13.50%	97.30%
	99.5%	2.21%	6.87%	84.30%
	99.9%	5.08%	14.84%	76.20%
Long USD/Short JPY	99.0%	7.37%	4.59%	23.30%
	99.5%	5.44%	0.00%	36.10%
	99.9%	7.87%	20.90%	54.70%
Short USD/Long JPY	99.0%	0.51%	1.10%	100.00%
	99.5%	1.23%	4.13%	100.00%
	99.9%	NA (no exceedances)	NA	NA (no exceedances)
3-day horizon				
Long AUD/Short USD	99.0%	12.65%	23.81%	99.40%
Short AUD/Long USD	99.0%	4.50%	13.02%	99.80%
Long GBP/Short USD	99.0%	7.61%	15.16%	89.30%
Short GBP/Long USD	99.0%	74.38%	74.67%	59.50%
Long USD/Short JPY	99.0%	0.62%	0.74%	3.90%
Short USD/Long JPY	99.0%	19.91%	39.26%	63.00%

- (a) The trading portfolio used for the analysis.
(b) The confidence level $(1-\alpha)$ at which VaR and ETL are calculated.
(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.
(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.
(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

Table 9: Backtest of Conditional Normal Mixture

Sample Size = 2000 days

Portfolio	Confidence Level	Unconditional Coverage VaR p-value	Conditional Coverage VaR p-value	ETL p-value
(a)	(b)	(c)	(d)	(e)
1-day horizon				
Long AUD/Short USD	99.0%	40.90%	56.93%	0.30%
	99.5%	9.71%	21.00%	1.10%
	99.9%	0.68%	2.50%	19.10%
Short AUD/Long USD	99.0%	1.24%	2.03%	5.80%
	99.5%	0.12%	0.38%	40.80%
	99.9%	27.03%	53.94%	8.40%
Long GBP/Short USD	99.0%	57.14%	41.65%	5.60%
	99.5%	35.69%	52.92%	2.70%
	99.9%	3.72%	11.15%	5.00%
Short GBP/Long USD	99.0%	7.37%	12.67%	0.90%
	99.5%	11.07%	21.57%	0.50%
	99.9%	0.68%	2.49%	0.10%
Long USD/Short JPY	99.0%	55.67%	0.00%	0.00%
	99.5%	7.84%	0.00%	0.00%
	99.9%	0.00%	0.00%	0.00%
Short USD/Long JPY	99.0%	64.79%	77.23%	1.10%
	99.5%	56.53%	35.18%	3.40%
	99.9%	3.72%	11.15%	61.90%
3-day horizon				
Long AUD/Short USD	99.0%	12.65%	23.81%	64.10%
Short AUD/Long USD	99.0%	49.57%	73.04%	97.10%
Long GBP/Short USD	99.0%	3.04%	7.59%	35.30%
Short GBP/Long USD	99.0%	24.21%	36.13%	0.60%
Long USD/Short JPY	99.0%	4.88%	9.19%	0.00%
Short USD/Long JPY	99.0%	73.44%	79.04%	3.30%

(a) The trading portfolio used for the analysis.

(b) The confidence level ($1-\alpha$) at which VaR and ETL are calculated.

(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.

(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.

(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

- Large samples (say 2000 days) are preferable to smaller samples (say 250 days) for risk estimation. This is especially true for the four conditional models and also for the unconditional empirical model. For the remaining three models, the choice of sample size appears less important. Space does not permit presentation of results for all sample sizes,¹² but we show the outcomes for the conditional t with small sample size in Table 10. Tables 8 and 10 are identical in every respect except for sample size. These findings are consistent with evidence by Lumsdaine (1995) with regard to estimating GARCH models.

12 Full details are available from the authors on request.

Table 10: Backtest of Conditional t

Sample Size = 250 days

Portfolio (a)	Confidence Level (b)	Unconditional Coverage VaR p-value (c)	Conditional Coverage VaR p-value (d)	ETL p-value (e)
1-day horizon				
Long AUD/Short USD	99.0%	0.21%	0.88%	0.50%
	99.5%	0.01%	0.03%	13.80%
	99.9%	1.97%	6.45%	1.60%
Short AUD/Long USD	99.0%	98.68%	68.73%	95.70%
	99.5%	55.84%	78.59%	97.70%
	99.9%	9.60%	25.02%	0.00%
Long GBP/Short USD	99.0%	23.63%	21.65%	53.30%
	99.5%	45.39%	61.83%	31.90%
	99.9%	28.30%	55.46%	27.10%
Short GBP/Long USD	99.0%	94.58%	89.35%	30.20%
	99.5%	68.97%	77.29%	14.80%
	99.9%	15.51%	35.81%	38.20%
Long USD/Short JPY	99.0%	0.00%	0.00%	0.30%
	99.5%	0.00%	0.00%	1.10%
	99.9%	0.01%	0.01%	1.20%
Short USD/Long JPY	99.0%	7.14%	13.67%	98.30%
	99.5%	6.11%	16.11%	93.20%
	99.9%	NA (no exceedances)	NA (no exceedances)	NA (no exceedances)
3-day horizon				
Long AUD/Short USD	99.0%	2.35%	5.34%	9.40%
Short AUD/Long USD	99.0%	4.50%	13.02%	80.40%
Long GBP/Short USD	99.0%	3.04%	7.59%	82.10%
Short GBP/Long USD	99.0%	74.38%	74.67%	24.60%
Long USD/Short JPY	99.0%	0.62%	2.05%	0.10%
Short USD/Long JPY	99.0%	42.22%	62.86%	28.40%

(a) The trading portfolio used for the analysis.

(b) The confidence level $(1-\alpha)$ at which VaR and ETL are calculated.

(c) Test of the null hypothesis that the actual number of violations is equal to the expected number of violations.

(d) Joint test of the null hypothesis that violations are spread evenly over time (as opposed to being clustered) and (c) above.

(e) Test of the null hypothesis that the ETL does not consistently understate the true potential for losses beyond the VaR.

To summarise the results of the empirical analysis, we find that currency analysts wishing to conduct a stress test in the context of a risk model should adopt risk models that are both conditional and allow for heavy-tails relative to the normal. We recommend either the conditional empirical approach or the conditional t , with the former being preferable if data are highly skewed. We also recommend that relatively large samples with at least 1000 observations are used to estimate the parameters of a risk model.

5. Evaluating Model Based Stress Tests

This section explains the basic parameters used in our stress tests and compares stress tests results obtained from different risk models with those obtained from more traditional stress tests. We also focus on the implications of model-based stress testing for capital adequacy.

5.1 Stress Testing Parameters

The empirical results in Section 4 show that out-of-sample predictions of extreme percentiles are most accurate when based on the Conditional Empirical or the Conditional t risk models. None of the unconditional risk models were very accurate, but from these the most suitable appeared to be the Unconditional Empirical. We shall therefore include this model as a benchmark for model-based stress testing, anticipating a superior result from the two conditional models. Taking these three risk models, stress tests are performed with the model-based method described in Section 3 using all available data from the sources described in Section 4.

There are three basic parameters for these tests:

- The size of the first extreme event, α : We have set $\alpha = 0.0002$, which is equivalent to an initial shock of a size that just breaches the 99.98% confidence level predicted by the risk model, and $\alpha = 0.0005$, i.e. an initial shock just beyond the 99.95% confidence level;
- The definition of ‘worst-case’ loss, p : We have chosen p equal to 0.01, which is equivalent to a one-sided 99% confidence interval for stress test outcomes. In other words, our results are reported as the 99% worst case loss occurring during a period of h days starting with the initial shock;
- The holding period for the test, h : This is the number of days that the portfolio is held, after the initial extreme event, before it is fully hedged against further losses.

Regarding the last parameter, we examined the stress tests results for risk horizons up to 20 days. Figures 1-3 present the results for the conditional empirical model only for three currency portfolios. For comparison, the regulatory capital computed under the current rules is also depicted on the graph (see explanation in Section 5.4 below). The risk horizon for the stress tests is on the horizontal axis and only the 99% worst-case loss is shown on the chart. We consider two possibilities for the initial shock, setting α equal to either 0.0002 or 0.0005 as explained above.

Figure 1: Comparison of Basel I Regulatory Capital and Stress Test Losses
Long AUD/USD Portfolio Conditional Empirical Model

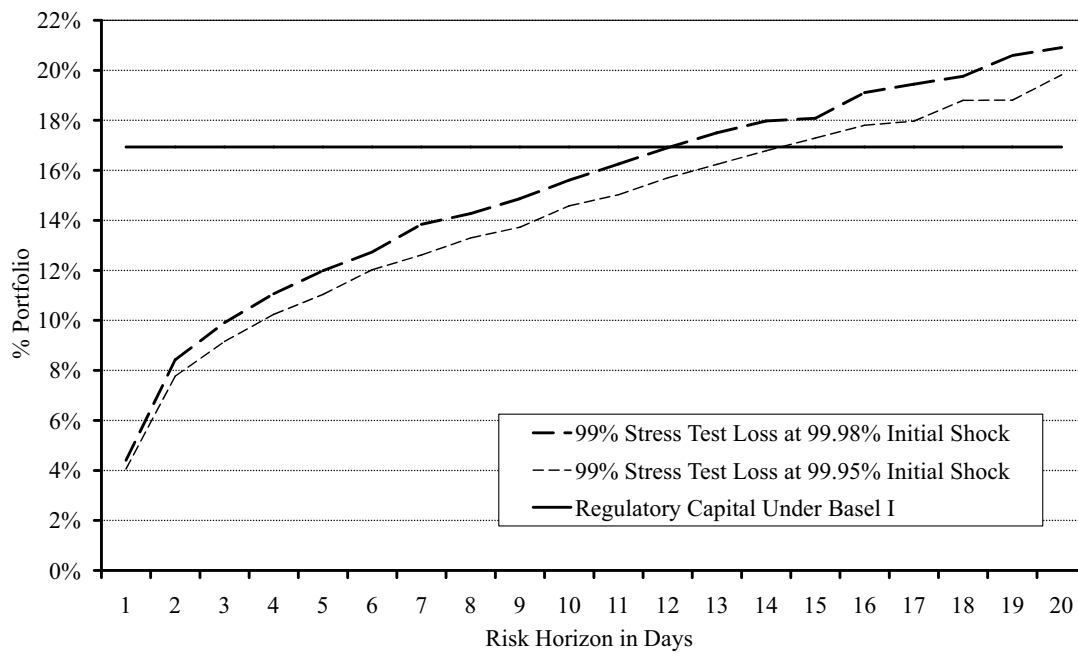


Figure 2: Comparison of Basel I Regulatory Capital and Stress Test Losses
Long GBP/USD Portfolio Conditional Empirical Model

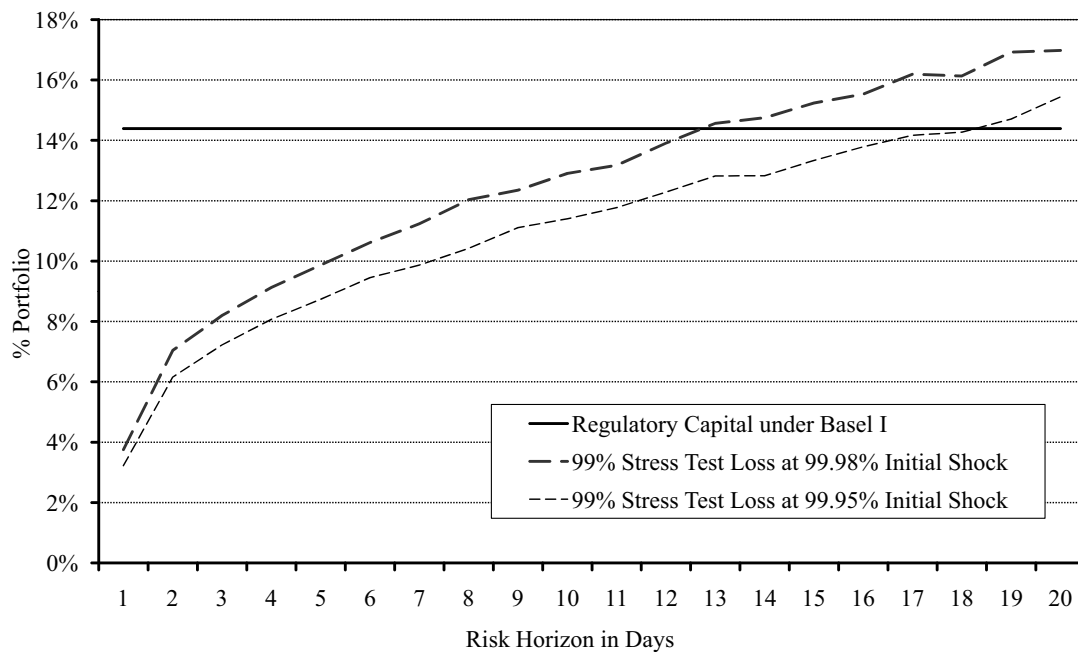
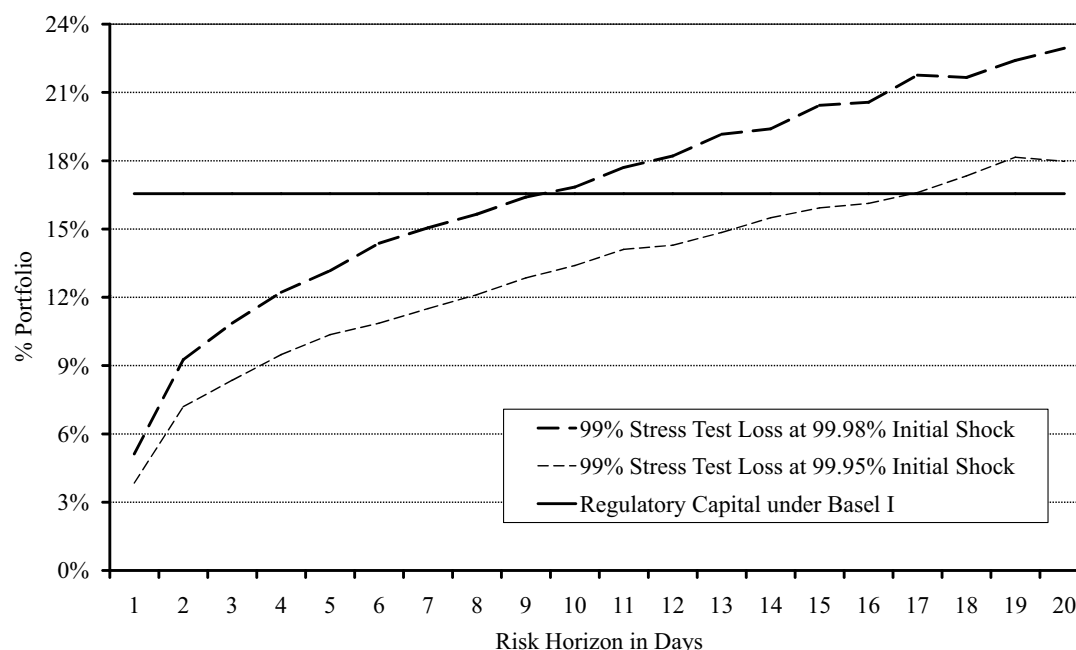


Figure 3: Comparison of Basel I Regulatory Capital and Stress Test Losses
Long USD/JPY Portfolio Conditional Empirical Model



Note that there is no regulatory prescription regarding the risk horizon of stress tests, in fact this is currently the subject of debate between regulators and the industry. The results displayed in Figure 1 indicate that Basel I risk capital exceeds the capital required to cover results of model-based stress tests at 99% worst case loss if the stress test horizon is not greater than about 12 or 14 days. The exact point where the stress test constraint starts to bite on regulatory capital calculations depends on the size of the initial shock. Of course, it depends on the portfolio and risk model too, as we shall see in the next section.

What single risk horizon, if any, should be applied to model-based stress tests? We sought the advice of some leading figures in the industry. Deutsche Bank in London use a minimum horizon of 10-days for stress-testing the very liquid forex positions.¹³ The GBP/USD falls into this category, and Figure 2 indicates that a 10-day stress test is unlikely to impinge on the existing regulatory capital calculated using the Basel I rules. For the USD/JPY, however, Figure 3 indicates that the losses from a 10-day stress test would marginally exceed current regulatory capital.

The choice of stress test horizon should account for both reduced market liquidity (which will vary by asset), the size of the position relative to the market and potential delays in managerial reaction to a shock event. But even the choice of a 10-day horizon may be

¹³ Longer horizons are used for less liquid assets, with a typical stress testing horizon being two months.

influenced by regulatory conservatism. A much shorter risk horizon (say, 3 days) may adequately reflect the time taken to hedge typical positions, considering that even the least liquid of the currency pairs we are examining (the AUD/USD) has average daily turnover of USD90 billion.¹⁴ The extent of risk is also reduced by the ability of financial institutions to undertake gradual liquidation over several hours/days. Gradual liquidation means that there is some diminution in risk almost immediately, with risk completely eliminated over time. In this context even a 3-day stress test builds in considerable conservatism, since it is equivalent in risk terms to gradual liquidation over a much longer horizon.

Figure 4: Stress Test Analysis¹⁵ Over Time

Long GBP/USD Portfolio, Conditional-t Model

99% Stress Test Loss with 99.98% Initial Shock, 10-day Horizon

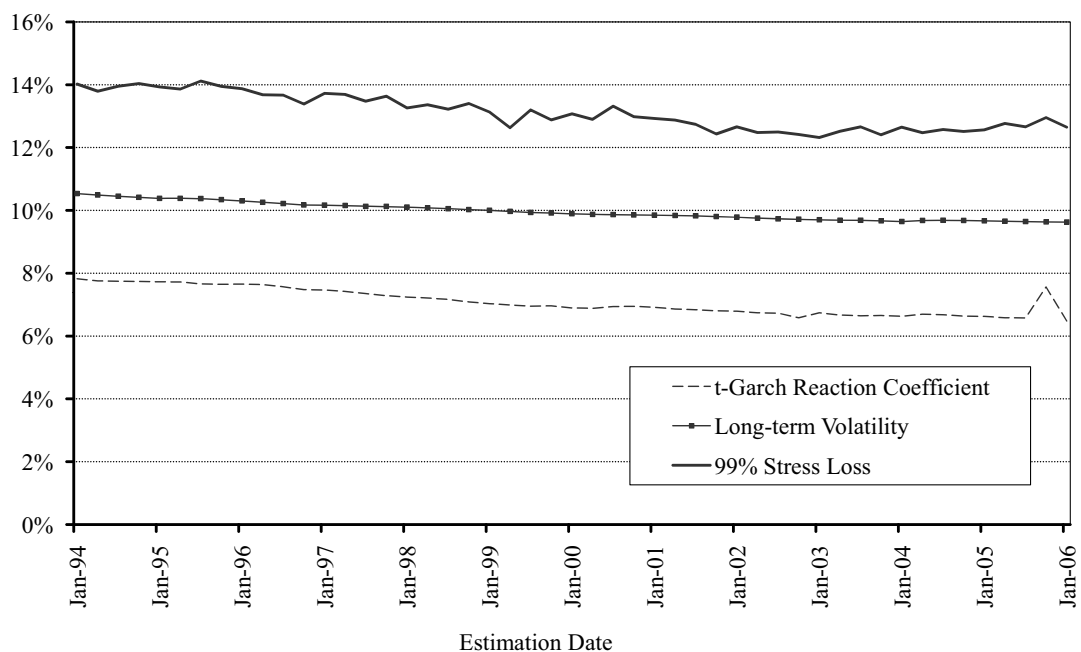


Figure 4 illustrates the stability of stress test parameters over time and hence the stress test results. Using the Long GBP/USD portfolio as a typical example, we first perform parameter estimation and the stress test analysis using 20 years of data from January 1974 to January 1984. Here we use the Conditional-t model with a 10-day horizon, an initial shock at 99.98% and the 99% worst case loss (i.e. the upper bound of the stress test is computed at the 99th percentile). This process is repeated at quarterly intervals, so that

¹⁴ Turnover figures were obtained from BIS (2005). In the USD/JPY average daily turnover is USD296 billion, while in GBP/USD average daily turnover is USD245 billion.

¹⁵ Analysis performed as at Estimation Date, using all available data from January 1974 until Estimation Date.

at each estimation point we use all available data from January 1974 until the estimation point. As the minimum sample period is 20 years, the parameter estimates are highly stable, for example, the reaction term from the GARCH equation (γ_2) is estimated in the range 0.065-0.078 and the degrees of freedom estimated for the t-GARCH model range between 4.6 and 5.6, gradually increasing over the sample. The 99% worst case loss is estimated in the range 12.3% to 14.1% of portfolio value, with a tendency to slightly decrease as the sample size increases. Thus the results we present using all available data (Figures 1-3 and Tables 11-13) can be viewed as relatively conservative.

5.2 Comparing Stress Tests by Risk Model

Our three preferred risk models may be compared by referring to Tables 11-13. These tables display the results for three of the six portfolios for risk horizons of 3 and 10 days (other portfolios/horizons are available from the authors on request). Consider, for example, the 3-day stress test performed on the long AUD portfolio with $\alpha = 0.0002$, presented in panel 1 of Table 11. The assumed initial shock to currency returns in row (*i*) varies between models from -4.41% (empirical) to -3.90% (conditional-*t*).¹⁶ The empirical models have a larger initial shock in this case because of the negative skew observed in AUD/USD returns.

A much larger discrepancy between the risk models is apparent when comparing the overall stress test outcomes over 3-day and 10-day horizons. The conditional models, incorporating volatility clustering, have a wider range of outcomes. The conditional empirical model has the widest range of all the risk models here, due to the asymmetry incorporated into the model. This model suggests that when a stress event occurs we are 99% confident that losses over 3 days will be no greater than 9.88% of portfolio value. The best possible outcome (with 99% confidence) is a loss of -0.002%. Notice that the potential for further losses after the shock is considerably greater than the potential for gains under the conditional empirical model. In contrast, the symmetric conditional-*t* model has very similar upside/downside potential.

5.3 Comparing Traditional Stress Tests with Model Based Stress Tests

According to the Committee on the Global Financial System (2005), the most common stress tests are based on historical events that would be potentially disastrous for the portfolio being analysed. For the currency portfolios in this study, we have identified the

¹⁶ We also calculated the initial shocks using EVT. We estimated ξ and β from the same empirical sample, and used these parameter estimates to calculate VaR_α , which can be used to determine the initial shock in the stress test. Calculating initial shocks in this way had negligible impact on the final stress test outcomes. Note that in every case we examined the estimated value of ξ was less than 0.1.

worst recorded currency movements from our data over horizons of 1, 3 and 10 days, as shown in the headers of Tables 11-13.

Table 11: Stress Test for Long AUD/Short USD Portfolio

Worst Case Historical Loss Over 1-day = 4.56%,

Over 3 days = 7.27%, Over 10 days = 12.61%

	Unconditional Empirical	Conditional Empirical	Conditional-t
a. Model Parameters	n/a	$\gamma_2 = 0.0659$ $\gamma_3 = 0.9243$	$\gamma_2 = 0.0652$ $\gamma_3 = 0.9262$ $v = 5.8314$
b. Long Term Volatility	n/a	10.36%	10.36%
c. VaR based Regulatory Capital	17.84%	16.93%	16.05%
1. 3 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	4.41%	4.41%	3.90%
ii) 99% Best Case Loss	2.10%	0.002%	(0.28%)
iii) 99% Worst Case Loss	7.07%	9.88%	8.03%
2. 3 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	4.06%	4.06%	3.26%
ii) 99% Best Case Loss	1.75%	0.02%	(0.34%)
iii) 99% Worst Case Loss	6.72%	9.07%	7.04%
3. 10 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	4.41%	4.41%	3.90%
ii) 99% Best Case Loss	(0.5%)	(4.67%)	(4.73%)
iii) 99% Worst Case Loss	10.05%	15.42%	12.46%
4. 10 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	4.06%	4.06%	3.26%
ii) 99% Best Case Loss	(0.85%)	(4.51%)	(4.31%)
iii) 99% Worst Case Loss	9.70%	14.54%	11.12%

- (a) Model parameters estimated from the specified risk model using all available log returns data from Dec 1983 to June 2006. Daily data is mean-corrected and annualised.
- (b) Long-term volatility estimated as sample standard deviation of all daily log returns, expressed on a per annum basis.
- (c) Regulatory capital is expressed as a percentage of the portfolio amount, calculated as $3\text{VaR}_{0.01,10\text{-day}}$ using the relevant risk model. VaR calculations use entire sample for calculating parameters/percentiles, and (where needed) the long-term volatility estimate from (b).
- (i) The assumed loss on day 1 of the stress test, with probability of α on any given day.
- (ii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the $(1 - \rho)$ percentile, with $h = 3$ or 10. Figures in parentheses refer to a profit.
- (iii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the ρ percentile, with $h = 3$ or 10.

Table 12: Stress Test for Long GBP/Short USD Portfolio

Worst Case Historical Loss Over 1-day = 3.84%,

Over 3 days = 7.45%, Over 10 days = 15.89%

	Unconditional Empirical	Conditional Empirical	Conditional-t
a. Model Parameters	n/a	$\gamma_2 = 0.0637$ $\gamma_3 = 0.9237$	$\gamma_2 = 0.0645$ $\gamma_3 = 0.9355$ $v = 5.5380$
b. Long Term Volatility	n/a	9.60%	9.60%
c. VaR based Regulatory Capital	15.94%	14.39%	13.97%
1. 3 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	3.75%	3.75%	3.61%
ii) 99% Best Case Loss	1.52%	(0.07%)	(0.15%)
iii) 99% Worst Case Loss	6.12%	8.20%	7.53%
2. 3 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	3.22%	3.22%	3.02%
ii) 99% Best Case Loss	0.99%	(0.16%)	(0.52%)
iii) 99% Worst Case Loss	5.60%	7.15%	6.37%
3. 10 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	3.75%	3.75%	3.61%
ii) 99% Best Case Loss	(0.97%)	(4.05%)	(4.56%)
iii) 99% Worst Case Loss	8.79%	12.84%	11.74%
4. 10 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	3.22%	3.22%	3.02%
ii) 99% Best Case Loss	(1.50%)	(4.06%)	(4.20%)
iii) 99% Worst Case Loss	8.26%	11.28%	10.26%

- (a) Model parameters estimated from the specified risk model using all available log returns data from Dec 1983 to June 2006. Daily data is mean-corrected and annualised.
- (b) Long-term volatility estimated as sample standard deviation of all daily log returns, expressed on a per annum basis.
- (c) Regulatory capital is expressed as a percentage of the portfolio amount, calculated as $3\text{VaR}_{0.01,10\text{-day}}$ using the relevant risk model. VaR calculations use entire sample for calculating parameters/percentiles, and (where needed) the long-term volatility estimate from (b).
- (i) The assumed loss on day 1 of the stress test, with probability of α on any given day.
- (ii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the $(1 - \rho)$ percentile, with $h = 3$ or 10. Figures in parentheses refer to a profit.
- (iii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the ρ percentile, with $h = 3$ or 10.

Table 13: Stress Test for Long USD/Short JPY Portfolio

Worst Case Historical Loss Over 1-day = 5.14%,

Over 3 days = 7.05%, Over 10 days = 12.77%

	Unconditional Empirical	Conditional Empirical	Conditional-t
a. Model Parameters	n/a	$\gamma_2 = 0.0526$ $\gamma_3 = 0.9456$	$\gamma_2 = 0.0483$ $\gamma_3 = 0.9517$ $v = 4.8008$
b. Long Term Volatility	n/a	10.44%	10.44%
c. VaR based Regulatory Capital	18.16%	16.55%	15.12%
1. 3 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	5.12%	5.12%	4.18%
ii) 99% Best Case Loss	2.76%	0.56%	0.12%
iii) 99% Worst Case Loss	7.82%	10.93%	8.33%
2. 3 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	3.84%	3.84%	3.45%
ii) 99% Best Case Loss	1.48%	0.15%	(0.26%)
iii) 99% Worst Case Loss	6.55%	8.47%	7.04%
3. 10 day Stress Test with $\alpha = 0.0002$, $\rho = 0.01$			
i) Initial Shock	5.12%	5.12%	4.18%
ii) 99% Best Case Loss	0.11%	(4.55%)	(4.16%)
iii) 99% Worst Case Loss	10.86%	17.06%	12.34%
4. 10 day Stress Test with $\alpha = 0.0005$, $\rho = 0.01$			
i) Initial Shock	3.84%	3.84%	3.45%
ii) 99% Best Case Loss	(1.17%)	(3.95%)	(3.93%)
iii) 99% Worst Case Loss	9.58%	13.19%	10.72%

- (a) Model parameters estimated from the specified risk model using all available log returns data from Dec 1983 to June 2006. Daily data is mean-corrected and annualised.
- (b) Long-term volatility estimated as sample standard deviation of all daily log returns, expressed on a per annum basis.
- (c) Regulatory capital is expressed as a percentage of the portfolio amount, calculated as $3\text{VaR}_{0.01,10\text{-day}}$ using the relevant risk model. VaR calculations use entire sample for calculating parameters/percentiles, and (where needed) the long-term volatility estimate from (b).
- (i) The assumed loss on day 1 of the stress test, with probability of α on any given day.
- (ii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the $(1 - \rho)$ percentile, with $h = 3$ or 10. Figures in parentheses refer to a profit.
- (iii) Refers to the loss over an h -day horizon, expressed as a percentage of initial portfolio value, at the ρ percentile, with $h = 3$ or 10.

Taking the Long AUD portfolio as an example, we note in Table 11 that the worst recorded 3-day loss was 7.27% of the portfolio value. We see that this outcome is within the range predicted under the conditional model based stress tests with $\alpha = 0.0002$ but not within the range predicted by the unconditional model. This is further evidence that conditional risk models are preferable for stress testing purposes.

So are model based stress tests superior to traditional stress tests based on historical extremes? Consider the Long AUD position, where the conditional- t risk model assumes an initial shock equivalent to an immediate loss of 3.90% of the portfolio value when $\alpha = 0.0002$. There have been only 4 days since the float of the AUD in 1983 on which losses have exceeded 3.90%, so when the stress test is limited to historical data it relies on a very

small sample indeed. In contrast the model based stress test can simulate any number of exchange rate paths which are different from, but consistent with, those experienced historically. A larger sample allows the analyst to form a far more reliable conclusion concerning the likely impact of the stress event.

A further advantage of the model based approach is that it avoids subjective assumptions regarding the implications of the stress event for the portfolio. The model based approach incorporates the characteristics of markets that have been established in research over more than twenty years to plot the likely consequences of a shock. These consequences can include further large movements in the same market (volatility clustering), large moves in other markets, increased correlation, increased implied volatility in option markets and reduced market liquidity.

The weakness of the model based approach is its vulnerability to model risks. These include the risk that we have selected an inappropriate model to describe past/current market dynamics, the uncertainty regarding parameter estimation and the risk that market dynamics may change significantly in the future. We have attempted to mitigate these risks with an extensive, tailored back testing procedure (Section 4) and by using long data sets for parameter estimation.

The methodology we have outlined relies on extensive data. While lengthy data sets are available in many markets, they are not available in all. In cases where markets have recently been deregulated, or where deregulation is considered likely in the near future, historical data may be available but not relevant. In these cases any stress test based on historical data (whether model based or not) will be problematic. One possible way forward would be to apply the risk model to proxy data from similar markets, such as those that have already experienced deregulation.

An example of model risk can be illustrated from our data with reference to the breakdown of the European Monetary System (EMS) in 1992. In September 1992 the GBP came under speculative attack as market participants anticipated the withdrawal of the GBP from the EMS and its subsequent depreciation. The greatest historical depreciation in the GBP/USD occurred at that time with a loss of 15.89% over a 10-day period. Note in Table 12 that this loss falls outside the bounds predicted by the model-based stress test (99% worst case loss of 12.84% over 10 days). Since the sharp depreciation in the GBP was widely anticipated, a hypothetical shock could have been applied to the stress test model. For example a hypothetical shock of -5.00%, which seems conservative in the circumstances, would have been sufficient to generate a 99% worst case loss of 17.40% over 10 days.

This example illustrates the need for risk managers to constantly monitor markets for potential structural breaks that even a sophisticated risk model cannot adequately predict. In these cases hypothetical shocks should be used as a complement to model-based shocks.

5.4 Implications of Model Based Stress Tests for Capital

We next assess the model based stress tests in the light of regulatory capital requirements. According to Basel Committee on Banking Supervision (2006), those regulated financial institutions using internal VaR models to assess capital are now required to hold sufficient capital to cover stress tests. This is in addition to the requirement that they have capital of at least:

$$\text{Max}[\text{Most recent estimate of } \text{VaR}_{10\text{-day}}, k \times \text{Average } \text{VaR}_{10\text{-day}} \text{ for past 60 days}]$$

where k is a multiplier determined by local regulators having a minimum value of 3.

In line (c) of Tables 11-13 we calculate regulatory capital as $3\text{VaR}_{0.01,10\text{-day}}$ using all available data for estimating risk model parameters/percentiles and, where necessary, using long-term sample volatility for sigma. So the regulatory capital estimates represent values typical of relatively benign market conditions.

Despite that, we observe that for reasonable risk horizons the regulatory capital requirements based on $\text{VaR}_{0.01,10\text{-day}}$ are almost always greater than the losses generated by model based stress tests. In the long AUD case (Table 11) with $\alpha = 0.0002$, the 10-day stress test gives a 99% worst case loss of 15.42%, which is 9% less than current regulatory capital of 16.93%. Applying a 3-day horizon we observe that the upper bound for the stress test (9.88%) is 42% lower than current regulatory capital.

In Figure 2 we observe that for long GBP/USD the risk horizon must extend to 13 days (with $\alpha = 0.0002$), or 19 days (with $\alpha = 0.0005$) before worst case losses from the stress test breach regulatory capital. While there is legitimate debate concerning the appropriate risk horizon for a stress test (see discussion in Section 5.1), it seems to us that it would be difficult to justify a 13 day horizon for stress testing in liquid markets such as these.

6. Conclusion

Stress testing is an established component of daily risk management for portfolios exposed to market risk. Indeed many financial institutions are required by regulators to implement stress tests and their importance is to be extended under the new capital accord.

Yet traditional stress tests can be criticised for being conducted outside the context of a risk model. The implications of this are that the probability of an extreme outcome is

unknown and many extreme yet plausible possibilities are ignored. In addition many stress tests fail to incorporate the characteristics that markets are known to exhibit in crisis periods, namely, increased probability of further large movements, increased co-movement between markets, greater implied volatility and reduced liquidity.

The first objective of this paper was to identify the most suitable risk model in which to conduct a stress test. We considered eight risk models, of which four are conditional on recent history, thus incorporating volatility clustering. We also explored the importance of heavy tails and skewness by comparing the normal, empirical, Student's *t* and normal mixture distributions. Analysis was performed using daily returns for the AUD/USD from December 1983 to June 2006 and for the GBP/USD and USD/JPY from January 1974 to June 2006. Our back testing procedure was designed to focus on extreme market movements over short risk horizons, as this is most relevant to stress testing. Conditional and unconditional coverage tests were performed at 99.0%, 99.5% and 99.9% for horizons up to 3 days using an extremely long out-of-sample returns series for each model. We also tested the accuracy of the expected tail loss estimates to ensure that our risk models adequately explain losses beyond VaR. We found that all risk models benefit from longer in-sample periods (in excess of 1000 days) and we concluded that the most suitable risk models for stress testing purposes should be conditional models with either an empirical or a Student's *t*-distribution (provided returns are not significantly skewed).

The second and major objective of this paper was to develop a methodology for conducting stress tests in the context of a risk model. We examined an initial shock event that was linked to the probability of its occurrence and modelled the consequences of that shock event using a risk model. We implemented this stress testing procedure for three major currency pairs and compared results with the traditional 'historical scenario' stress testing approach. Unconditional risk models seem unable to predict historical worst-case losses. However all stress tests based on conditional risk models could predict the worst case historical scenarios, the one exception being the Black Wednesday crash of the ERM that affected the GBP/USD rate. In cases such as this, where stress conditions are largely anticipated, we recommend that hypothetical rather than model-based initial shocks be applied to the risk model.

Finally, the implications for capital adequacy under the new Basel recommendations have been explored. A detailed comparison has indicated that current regulatory requirements for trading portfolios in liquid currency markets are already conservative enough to cover the results of stress tests. Hence no additional risk capital should be required under the new Basel recommendations.

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Appendices

1: Derivation of Analytic Formulae for ETL

If X has density $f(x)$ then

$$ETL_{\alpha}(X) = -\alpha^{-1} \int_{-\infty}^{X_{\alpha}} x f(x) dx \quad (\text{A.1})$$

We prove the following:

(i) If $X \sim N(\mu, \sigma^2)$ then

$$ETL_{\alpha}(X) = \alpha^{-1} \varphi(Z_{\alpha}) \sigma - \mu \quad (\text{A.2})$$

(ii) If $X \sim t_{\nu}(\mu, \sigma^2)$ has density $f_{\nu}(x)$ then

$$ETL_{\alpha, \nu} = \alpha^{-1} (\nu - 1)^{-1} (\nu - 2 + X_{\alpha}^2) f_{\nu}(X_{\alpha}) \sigma - \mu \quad (\text{A.3})$$

where $X_{\alpha} = -VaR_{\alpha, \nu} | t_{\nu}(0, 1)$

(iii) If $X \sim NM(\pi, \mu, \sigma^2)$ where

$$\pi = (\pi_1, \dots, \pi_{\nu}), \mu = (\mu_1, \dots, \mu_{\nu}) \text{ and } \sigma^2 = (\sigma_1^2, \dots, \sigma_{\nu}^2)$$

then

$$ETL_{\alpha}(X) = \alpha^{-1} \sum_{i=1}^n \left(\pi_i \varphi(\sigma_i^{-1} X_{\alpha}) \sigma_i \right) - \sum_{i=1}^n \pi_i \mu_i \quad (\text{A.4})$$

where $X_{\alpha} = -VaR_{\alpha} | NM(\pi, 0, \sigma^2)$

Proof:

(i) For a standard normal variable Z :

$$ETL_{\alpha}(Z) = -\alpha^{-1} \int_{-\infty}^{Z_{\alpha}} z \varphi(z) dz = -\left(\sqrt{2\pi} \alpha\right)^{-1} \int_{-\infty}^{Z_{\alpha}} z \exp(-0.5z^2) dz = \alpha^{-1} \varphi(Z_{\alpha})$$

Write:

$$X = Z\sigma + \mu, \quad Z \sim N(0, 1)$$

and now (A.2) follows from (A.1).

(ii) The result follows on showing that the ETL in a standardized Student's t-distribution $t_{\nu}(0, 1)$ with ν degrees of freedom, unit standard deviation and zero mean is given by:

$$\alpha^{-1} (\nu - 1)^{-1} (\nu - 2 + X_\alpha^2) f_\nu (X_\alpha) \quad (\text{A.5})$$

The density function for $t_\nu (0,1)$ is:

$$f_\nu (x) = ((\nu - 2) \pi)^{-1/2} \Gamma\left(\frac{\nu}{2}\right)^{-1} \Gamma\left(\frac{\nu + 1}{2}\right) (1 + (\nu - 2)^{-1} x^2)^{-\left(\frac{\nu + 1}{2}\right)}$$

Write:

$$\begin{aligned} f_\nu (x) &= A (1 + ax^2)^b, \\ A &= (\beta \nu \pi)^{-1/2} \Gamma\left(\frac{\nu}{2}\right)^{-1} \Gamma\left(\frac{\nu + 1}{2}\right), \\ a &= (\nu - 2)^{-1}, \text{ and} \\ b &= -\left(\frac{\nu + 1}{2}\right) \end{aligned}$$

Then:

$$\int_{-\infty}^{X_\alpha} x f_\nu (x) dx = A \int_{-\infty}^{X_\alpha} x (1 + ax^2)^b dx = \frac{A}{2a} \int_{-\infty}^{Y_\alpha} y^b dy, \quad Y_\alpha = 1 + \frac{X_\alpha^2}{\nu - 2}$$

on making the substitution $y = 1 + ax^2$. The integral becomes:

$$\int_{-\infty}^{Y_\alpha} y^b dy = \frac{Y_\alpha^{b+1}}{b+1} = -Y_\alpha^{-\left(\frac{\nu-1}{2}\right)} \left(\frac{\nu-1}{2}\right)^{-1}$$

Thus in the standardized t-distribution

$$ETL_{\alpha, \nu} = \alpha^{-1} A \frac{\nu - 2}{\nu - 1} Y_\alpha^{-\frac{(\nu-1)}{2}} = \alpha^{-1} \frac{A}{\nu - 1} (\nu - 2 + X_\alpha^2)^{-\frac{(\nu-1)}{2}} \quad (\text{A.6})$$

But

$$A = \left(1 + \frac{X_\alpha^2}{\nu - 2}\right)^{\frac{(\nu+1)}{2}} f_\nu (X_\alpha)$$

and substituting this in (A.6) gives (A.5).

(iii) First suppose X has a normal mixture with zero means, and write the density:

$$\eta (x) = \sum_{i=1}^n \pi_i f_i (x)$$

where each $f_i (x)$ is a zero mean normal density with standard deviation σ_i . Thus

$$f_i(x) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-0.5\left(\frac{x}{\sigma_i}\right)^2\right)$$

and so by (A.1):

$$ETL_\alpha(X) = -\alpha^{-1} \int_{-\infty}^{X_\alpha} x \eta(x) dx = -\alpha^{-1} \sum_{i=1}^n \pi_i \int_{-\infty}^{X_\alpha} x f_i(x) dx$$

where $X_\alpha = -VaR_\alpha \mid NM(\pi, 0, \sigma^2)$.

For any normal density $f(x)$, with mean zero and standard deviation sigma:

$$\int_{-\infty}^{X_\alpha} x f(x) dx = -\frac{\sigma}{\sqrt{2\pi}} \exp\left[-0.5\left(\frac{X_\alpha}{\sigma}\right)^2\right] = -\sigma \varphi(\sigma^{-1} X_\alpha)$$

Hence for the zero-mean normal mixture:

$$ETL_\alpha(X) = \alpha^{-1} \sum_{i=1}^n \left(\pi_i \sigma_i \varphi(\sigma_i^{-1} X_\alpha) \right)$$

and the last term in (A.4) follows immediately, noting that $\sum_{i=1}^n \pi_i \mu_i$ is the mean of the normal mixture.

2: Scaling Empirical VaR and ETL

Scaling rules for percentiles can be derived when the portfolio returns have a *stable distribution*. When a distribution is ξ -stable then the whole distribution, including the percentiles, scales as $h^{1/\xi}$. In a normal distribution $\xi = 2$ and so its scale exponent is $1/2$. For other stable distributions ξ must lie in the range $(0, 2]$. Hence scaling by the square root of time will lead to an underestimation of the percentile of h -day returns, and hence the Value-at-Risk, when $0 < \xi < 2$ (see Danielsson and Zigrand, 2006).

We can estimate the *scale exponent* ξ^{-1} for scaling the distribution of a log return X and for scaling its percentiles in particular. Let $X_{h,\alpha}$ denote the lower α percentile of the h -day discounted log returns. We seek ξ such that:

$$X_{h,\alpha} = h^{1/\xi} X_{1,\alpha} \quad (\text{A.7})$$

In other words:

$$\xi = \frac{\ln(h)}{\ln(X_{h,\alpha}) - \ln(X_{1,\alpha})}$$

Hence ξ can be estimated as the slope of graph with $\ln(X_{h,\alpha}) - \ln(X_{1,\alpha})$ on the horizontal axis and $\ln(h)$ on the vertical axis. If the distribution is stable the graph will be a straight line and ξ will not depend on the choice of α . Nor should it vary much when different samples are used, provided the sample contains sufficient data to estimate the percentiles accurately. When a constant scale exponent corresponding to (A.7) exists, we say that the log return obeys a *power law* scaling rule with exponent ξ^{-1} .

Figure 5: Log-Log Plot for the Power Law Scale Exponent, GBP/USD

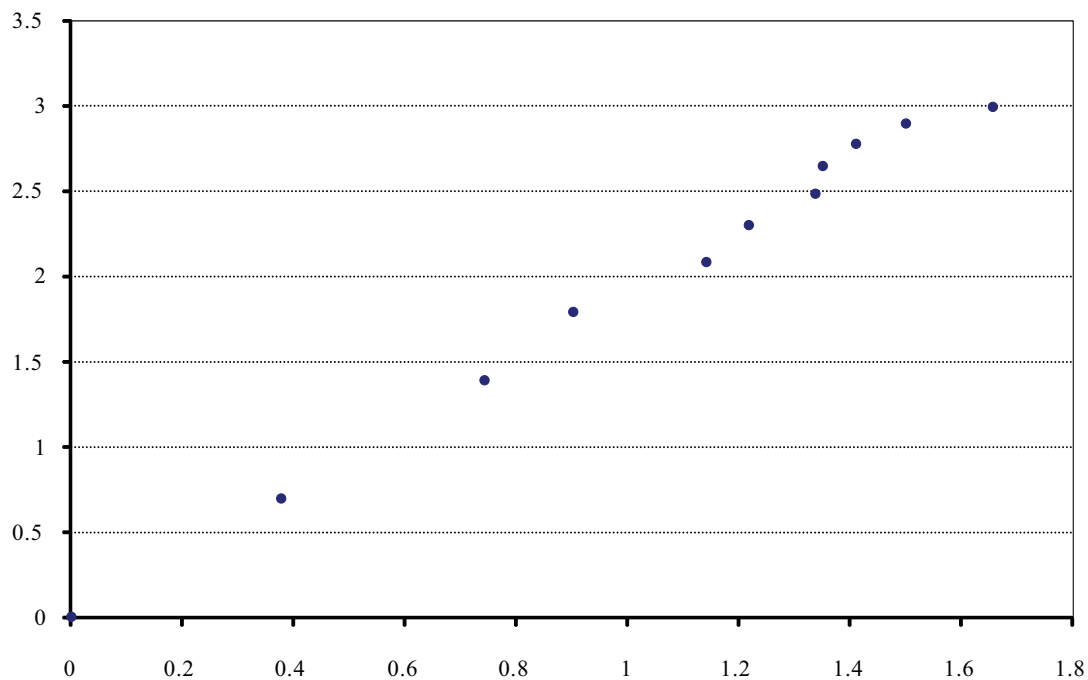


Figure 5 shows the log-log plot for the GBP rate. The scale exponent is the reciprocal of slope and with $\alpha = 5\%$, we have $\xi^{-1} = 0.5298$. A similar calculation for the AUD rate gives the scale exponent 0.5220. The estimated scale exponents appear to decrease marginally as we move to more extreme percentiles. For instance in the GBP rate with $\alpha = 1\%$, we obtain $\xi^{-1} = 0.4983$, indicating that the distribution is not strictly stable. But note that the estimation error increases as α decreases. At low percentiles estimation becomes imprecise due to lack of historical data. With $\alpha = 5\%$ all data give log-log plots such as Figure 5 that are nearly straight lines with slope $\approx \frac{1}{2}$ and therefore our empirical investigation indicates that the square root of time rule is a suitable scaling law for the percentiles.

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